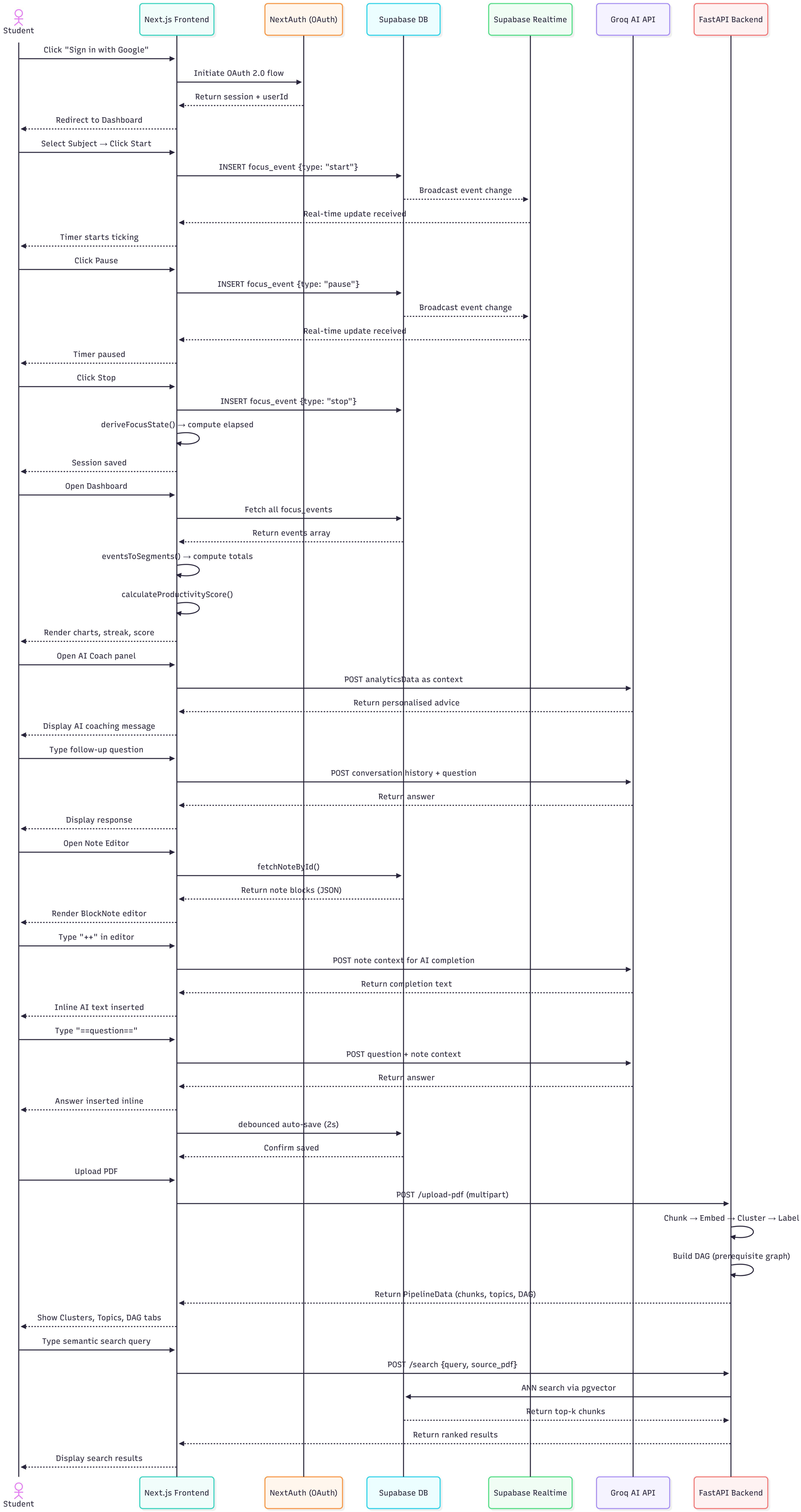
Student
Next.js Frontend
NextAuth (OAuth)
Supabase DB
Supabase Realtime
Groq AI API
FastAPI Backend
Click "Sign in with Google"
Initiate OAuth 2.0 flow
Return session + userId
Redirect to Dashboard
Select Subject → Click Start
INSERT focus_event {type: "start"}
Broadcast event change
Real-time update received
Timer starts ticking
Click Pause
INSERT focus_event {type: "pause"}
Broadcast event change
Real-time update received
Timer paused
Click Stop
INSERT focus_event {type: "stop"}
deriveFocusState() → compute elapsed
Session saved
Open Dashboard
Fetch all focus_events
Return events array
eventsToSegments() → compute totals
calculateProductivityScore()
Render charts, streak, score
Open AI Coach panel
POST analyticsData as context
Return personalised advice
Display AI coaching message
Type follow-up question
POST conversation history + question
Return answer
Display response
Open Note Editor
fetchNoteById()
Return note blocks (JSON)
Render BlockNote editor
Type "++" in editor
POST note context for AI completion
Return completion text
Inline AI text inserted
Type "==question=="
POST question + note context
Return answer
Answer inserted inline
debounced auto-save (2s)
Confirm saved
Upload PDF
POST /upload-pdf (multipart)
Chunk → Embed → Cluster → Label
Build DAG (prerequisite graph)
Return PipelineData (chunks, topics, DAG)
Show Clusters, Topics, DAG tabs
Type semantic search query
POST /search {query, source_pdf}
ANN search via pgvector
Return top-k chunks
Return ranked results
Display search results
Student
Next.js Frontend
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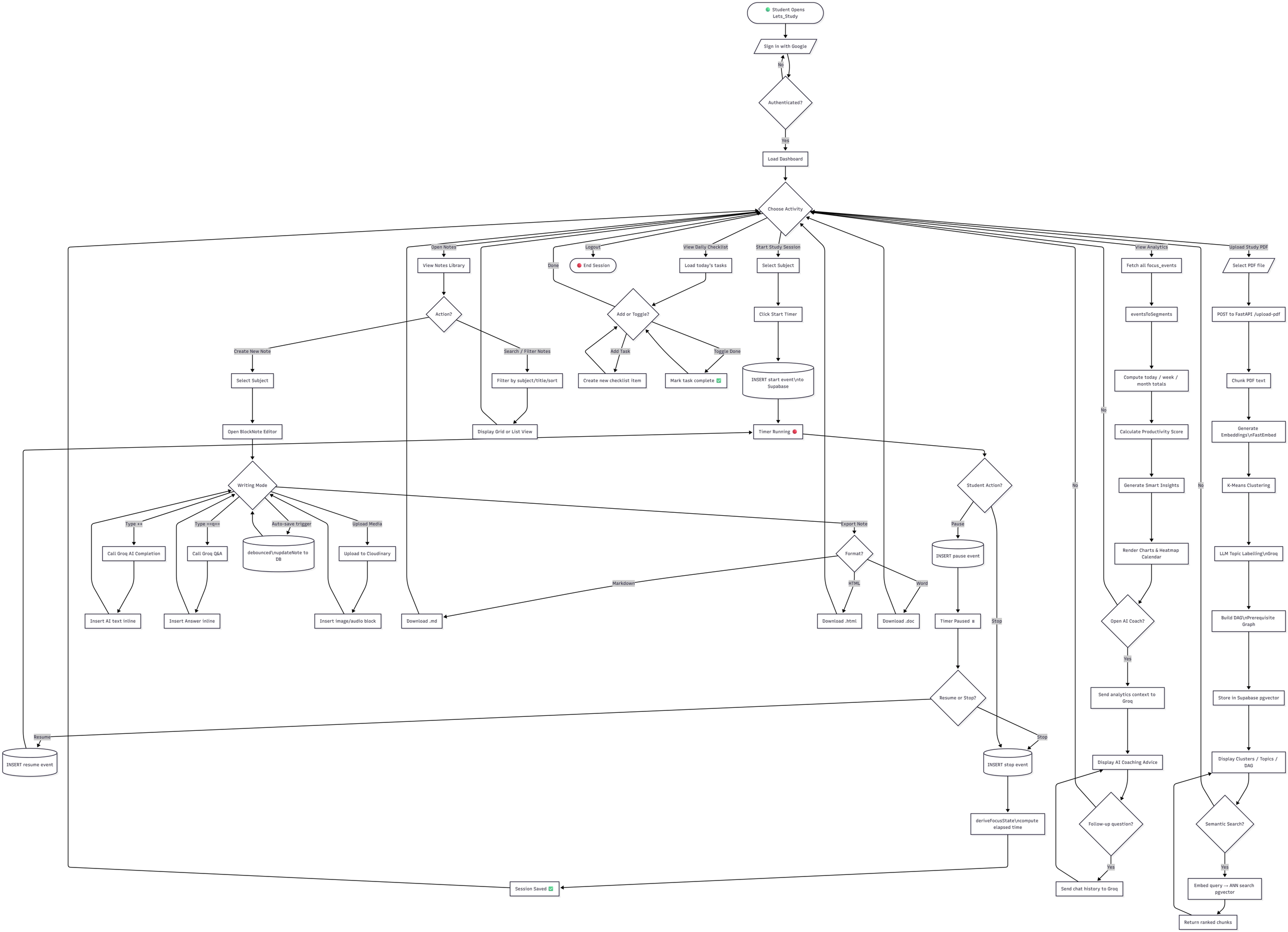
Student Opens Lets_Study
Sign in with Google
No
Authenticated?
Yes
Load Dashboard
Choose Activity
Open Notes
View Notes Library
Action?
Create New Note
Select Subject
Open BlockNote Editor
Search / Filter Notes
Filter by subject/title/sort
Display Grid or List View
Logout
End Session
Done
View Daily Checklist
Load today's tasks
Add or Toggle?
Add Task
Create new checklist item
Toggle Done
Mark task complete ✅
Start Study Session
Select Subject
Click Start Timer
INSERT start event\nto Supabase
Timer Running 🔴
Student Action?
Pause
INSERT pause event
Timer Paused ⏸
Resume or Stop?
Resume
INSERT resume event
Stop
INSERT stop event
deriveFocusState\ncompute elapsed time
Session Saved ✅
Writing Mode
Type ++
Call Groq AI Completion
Insert AI text inline
Type ==q==
Call Groq Q&A
Insert Answer inline
Auto-save trigger
debounced\nupdateNote to DB
Upload Media
Upload to Cloudinary
Insert image/audio block
Export Note
Format?
Markdown
Download .md
HTML
Download .html
Word
Download .doc
View Analytics
Fetch all focus_events
eventsToSegments
Compute today / week / month totals
Calculate Productivity Score
Generate Smart Insights
Render Charts & Heatmap Calendar
Open AI Coach?
Yes
Send analytics context to Groq
Display AI Coaching Advice
Follow-up question?
Yes
Send chat history to Groq
No
Upload Study PDF
Select PDF file
POST to FastAPI /upload-pdf
Chunk PDF text
Generate Embeddings\nFastEmbed
K-Means Clustering
LLM Topic Labelling\nGroq
Build DAG\nPrerequisite Graph
Store in Supabase pgvector
Display Clusters / Topics / DAG
Semantic Search?
Yes
Embed query → ANN search pgvector
Return ranked chunks
No